

ADITYA P DIXIT

AI/ML Engineering Student | Open Source Systems Builder

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PROFILE

AI/ML engineering student focused on autonomous systems, AI infrastructure, and developer tooling. Experienced in building agent frameworks, reinforcement learning systems, and open-source AI tooling using Node.js, Python, and PyTorch.

EDUCATION

B.Tech (Hons.) - Artificial Intelligence & Machine Learning, RV University
CGPA: 9.00
Major: Artificial Intelligence and Machine Learning
Minor: Film Direction

08/2024 – Present
Bengaluru

SKILLS

Languages — Python, JavaScript, Java, Dart, SQL

Frameworks & Libraries — Node.js, Flutter, PyTorch, ML Kit, Pandas, NumPy, Matplotlib, Scikit-learn, Statistics

AI/ML — LLMs, Reinforcement Learning, MCP, RAG Systems, LLM Safety

Tooling & Infrastructure — Git, GitHub Actions, OpenAPI, CI/CD, Docker, Kubernetes

Soft Skills — Analytical Thinking, Attention to Detail, Cross-functional Collaboration, Time Management

ACHIEVEMENTS & AWARDS

1st Place Kalpavikas 1.0 Ideathon Inter-College, 95+ Competing Teams, RV University
Won first place among 95+ competing teams in an inter-collegiate ideathon, demonstrating product thinking, technical depth, and strong presentation under competitive conditions.

LEADERSHIP & EXPERIENCE

Student Assistantship, Library By brigade, RV University
• Managed 40+ hours/week of operational and administrative workflows supporting university library systems and staff coordination.
• Assisted the library in multiple domains including AI-assisted Workload management and using LLMs to complete tasks faster.

03/2026 – 08/2026
Bengaluru

Student Placement Coordinator, RV University
• One of the student coordinators managing university placement operations and coordination workflows.

03/2026 – Present
Bengaluru

Open Source Head, FOSS United
• FOSS United RVU Lead open-source contribution reviews, test builds, and validate functionality before merges mirroring professional code-review and QA workflows.

01/2026 – Present
Bengaluru

WebOps Head, Iterium
• Manage the club's web presence, tooling, and digital operations

10/2025 – Present
Bengaluru

Event Management Head, Vyoma
Vyoma Space Club, RV University Coordinate planning, logistics, and execution of technical events and initiatives; identify failure points and ensure smooth delivery.

10/2025 – Present
Bengaluru

Head of Literature, Poéthra
• Led planning and execution of literary events; part of the organising committee for Brigade's Book Festival 2025.

08/2025 – Present
Bengaluru

TECHNICAL PROJECTS

RexScript — Experimental Language & Runtime for Autonomous Systems 
Compiler Design, Tooling, Runtime Systems
• Designed and implemented an experimental scripting language for autonomous agent workflows with custom execution semantics and runtime tooling.
• Built compiler-backed diagnostics, execution tracing, and multi-mode runtime support with integrated VS Code tooling and syntax highlighting.
• Curated developer tooling including command workflows, editor integrations, and structured execution pipelines for .rex programs.

05/2026 – Present

XRisk Engine - Open-Source AI Safety & Control Layer 
• **Engineered** a policy-enforcement layer for LLM agents supporting **3 decision states (allow / confirm / block)** with a normalized **0–1 risk scoring system**
• Created a **6-signal weighted risk model** with defined thresholds (**allow <0.65, block ≥0.85**) for consistent decision-making

02/2025 – Present

- Implemented prompt-injection defense (**6 detection patterns**) and sensitive data protection across **5 data classes** (API keys, PII, etc.)
 - Recorded **100% pass rate across 20+ scenario tests** with full security and core validation suites
- DQL Agent - Deep Reinforcement Learning**, *PyTorch* 04/2026 – 05/2026
- Trained Deep Q-learning agents to solve Pac-Man, Minesweeper, and Snake using custom reward system and experience replay
 - Achieved successful completion across all 20 Pac-Man evaluation episodes with average reward of 58.62
 - Improved training performance by **+5.95 reward** across episodes through iterative tuning
- Firefly — Deterministic Systems Diagnostics Platform** *⌘, Tauri, Node.js, Systems Modeling* 05/2026 – Present
- Built a topology-aware diagnostics platform for modeling expected vs observed system performance across storage, PCIe, memory, and network paths.
 - Executed deterministic bottleneck analysis, hardware probing, benchmark tooling, and cross-platform desktop packaging using Tauri.
 - Developed local-first benchmarking and diagnostics workflows with visual regression testing and CI automation.
- Intents Cognitive Continuity Browser Extension** *⌘* 01/2026 – 05/2026
- Built a local-first browser extension designed to help users preserve and recover browsing intent after distraction, replacing the new tab with a calm, minimal surface and maintaining a lightweight context trail through sessions.
 - Key features: Hold That Thought (save selected text with context and return to the exact page and highlight later), a Context Ladder (short-term session navigation trail), Isolate/reading mode, and an in-page Contextualise panel for selected-text explanations without opening new tabs.
 - Entirely local-first with no accounts, analytics, tracking, or telemetry;
 - designed around privacy and inspectability with open-source code.

CERTIFICATIONS

Introduction to Model Context Protocol Anthropic Academy	AWS Web Fundamentals Amazon Web Services	AI Fluency for Students Anthropic Academy	SQL (Advanced) Hackerrank
Digital Marketing Certification hubSpot Academy			